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A machine, not an intelligence: how Chinese programmers imagine generative artificial intelligence

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ABSTRACT

Imaginaries have become an essential analytic lens for understanding how emerging technologies like generative artificial intelligence (AI) are envisioned. While existing studies have begun to explore these collective visions, they remain predominantly focused on a macro-level perspective, emphasizing national and public imaginaries, while overlooking those shaped by key professional groups. This study addresses this gap by examining how programmers, who occupy an intermediary role between generative AI and society, imagine this transformative technology. Drawing on insights from the Social Construction of Technology (SCOT) and the concept of imaginaries, we adopt a mixed-methods approach combining computational text analysis of online forum posts with a national survey of 14,445 Chinese programmers. Online discussions reveal two imaginaries: "AI-as-Machine," which frames AI as a powerful yet controllable tool, and "AI-as-Intelligence," which casts it as an autonomous and potentially disruptive agent. Survey evidence further confirms the dominance of the "AI-as-Machine" imaginaries, particularly among technically proficient programmers. By centering programmers' pivotal position, this study not only grounds imaginaries in concrete metaphors, but offers a micro-foundation for understanding China's rapid, tool-driven adoption of generative AI.

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Introduction

Sociologists of technology have long argued that the development and social consequences of emerging technologies are not solely determined by inventors or scientific elites. Rather, they are co-constructed through the practices and interpretations of the wider public who incorporate these technologies into everyday life (Williams and Edge 1996). Within this tradition, the Social Construction of Technology (SCOT) framework has gained prominence for highlighting the interpretative flexibility of technology (MacKenzie and Wajcman 1999, 21–23). According to SCOT, technological artifacts may acquire divergent meanings depending on the social groups involved in the process of design, implementation, and use. These meanings, in turn, produce different social outcomes (Pinch and Bijker 1984).

In the case of generative artificial intelligence (AI), determinist narratives persist in subtle and residual forms (Wachter-Boettcher 2018, 9), despite long-standing critiques in SCOT and the wider sociology of technology (Joyce et al. 2023). Such narratives often appear in exaggerated claims about novelty and in portrayals of generative AI as an external force disrupting economic and social systems. While it is true that generative AI can now perform tasks involving cognition and abstraction (Du et al. 2025), thereby shifting task boundaries and potentially challenging occupational structures (Acemoglu et al. 2022), these capabilities are always mediated through social practices. A recent OpenAI-supported study shows that ChatGPT is adopted for diverse purposes, from coding and debugging to report writing, language learning, and everyday information search (Chatterji et al. 2025). Crucially, these practices vary systematically across educational backgrounds, professions, and industries, indicating that imaginaries of generative AI are already stratified along social lines. As a result, a sociological approach must begin by examining how these capabilities are interpreted and acted upon by different social groups.

Building on SCOT, more recent scholarship has introduced the concept of imaginaries to capture how collective visions of technology are both shaped by and constitutive of broader socioeconomic arrangements (Jasanoff and Kim 2009; McNeil et al. 2017; Richter, Katzenbach, and Zeng 2025). Unlike SCOT, which usually focuses retrospectively on processes of closure and stabilization, imaginaries are especially well suited for analyzing forward-looking and anticipatory orientations. This distinction is especially salient for generative AI, whose meaning is still being negotiated with reference to futures not yet realized.

At the same time, imaginaries concern not only cultural perceptions but also valuation processes. As Wang (2021) argues, in the digital era,

entrepreneurial value is increasingly generated through externalities and productive consumption among the masses, with collective narratives about technology shaping markets and directing investment flows. Recent media reports echo this dynamic: analysts caution that soaring valuations of AI startups are driven as much by investor sentiment and societal expectations as by actual revenues or technical performance (Ngui 2025; Shefrin 2025). These cases underscore how imaginaries can translate into market optimism, speculative bubbles, and strategic bets. Thus, imaginaries not only enrich theoretical debates in the science and technology studies (STS) but also illuminate pressing real-world dynamics.

Despite the importance of imaginaries in understanding emerging technologies, existing studies have applied the concept primarily to policymakers and publics (Selwyn and Gallo Cordoba 2022; Kuai 2025; Van Noordt, Medaglia, and Tangi 2025). The perspectives of professional groups, by contrast, remain notably absent. This article contributes to filling this gap by focusing on programmers, a key professional group positioned at an intermediary juncture between upstream innovation and downstream application, who play a pivotal role in shaping how AI technologies are socially embedded (Kotti, Galanopoulou, and Spinellis 2023).[1]

Moreover, this gap is particularly pronounced in the Chinese context, where the country has emerged as a global hub for generative AI, hosting more than 4,500 AI companies and receiving extensive state-led investment (Bicker 2025). China also supports a programmer population exceeding 9.4 million, ranking among the top three worldwide (GitHub Staff 2024; Sun et al., 2024). Researchers have also found that many Chinese programmers view learning and understanding AI as essential to their career advancement (Jia 2022), underscoring the urgency for closer analysis of this group's imaginaries.

Building on this focus on programmers, we adopt a mixed-methods approach grounded in empirical evidence from China, analyzed through the conceptual lens of imaginaries. First, we conduct a computational analysis of a large corpus of online forum posts to examine how Chinese programmers interpret and engage with generative AI. We then draw a nationwide survey to investigate how these imaginaries are distributed across the programming community, with particular attention to variations among highly skilled programmers. Understanding this intra-group heterogeneity is crucial, since those with greater technical expertise and organizational authority may play decisive roles in either facilitating adoption or expressing resistance (Burrell and Fourcade 2021; Campero 2021; Yan and Wang 2025).

The remainder of the article proceeds as follows. We first introduce SCOT as a theoretical foundation, followed by imaginaries as an analytical tool for examining programmers' anticipatory visions of generative AI.

Next, we review recent studies on AI imaginaries in China across national, public, and professional domains. We then describe our mixed-methods design and data sources, present our empirical findings, and conclude with a discussion of programmers' occupational imaginaries and their implications for understanding the rapid development of the China's AI industry.

Literature review

SCOT and its core concepts

Technology has never been peripheral to sociological thought. Sociology as a discipline emerged in response to the disruptions of industrialization, as an intellectual effort to grapple with the transformative power of technological change on labor, institutions, and social life (Giddens 1976). From its inception, classical sociological theorists have viewed technology not merely as an external instrument, but as a constitutive force in the making of modern society, an artifact shaping and shaped by human intention and social relations (Durkheim 1960; Marx 1982).

In dialogue with this macro-level tradition, SCOT redirected attention to the micro-dynamics of meaning-making and became a foundational framework in STS, offering a systematic approach to shift focus away from technological determinism and toward the social process of technological practice (MacKenzie and Wajcman 1999, 21). SCOT begins with the notion of interpretive flexibility, an idea that a single technological artifact can have multiple meanings and uses depending on how relevant social groups interpret it. Within each group, these interpretations are guided by a shared structure of goals, problem definitions, cognitive habits, and material conditions, collectively termed the technological frame (Bijker 1997, 124). As groups interact and negotiate meanings and uses, the range of interpretations narrows, leading the artifact toward closure and eventually stabilization (Pinch and Bijker 1984). This challenges earlier deterministic views that treat stabilized technologies as inevitable outcomes of technical features, rather than products of contingent social processes (Bijker and Bijsterveld 2000; MacKenzie and Wajcman 1999).

Through its foundational stance and key notions, SCOT offers a valuable starting point for understanding how technologies are socially embedded. As such, it remains a widely adopted framework for examining the evolving sociotechnical landscape whenever new technologies emerge. In this study, we aim to examine the programmers' "technological frame" around generative AI. However, as will be discussed below, the concept of imaginaries is better suited for capturing the forward-looking and interpretive dimensions of programmers' understanding of AI.

Beyond SCOT: imaginaries as a framework

While SCOT has been instrumental in foregrounding how technologies are socially constructed, critics have pointed out that it overlooked structural asymmetries by portraying relevant social groups as internally coherent and equally positioned, thereby obscuring the unequal power relations that affect technological outcomes (Klein and Kleinman 2002). As Hughes (1987) noted, technological systems are embedded in broader political, economic, and cultural contexts that extend beyond local negotiations. The assumption of equal participation risks masking the disproportionate influence of elite actors. Building on this critique, this study treats programmers not as a homogeneous group, but as an internally stratified community, where variations in technical expertise critically shape their power to influence technological imaginaries (Burrell and Fourcade 2021).

Apart from its limited engagement with structural asymmetries, SCOT also lacks robust analytical tools for tracing how the interpretations of a given technology evolve over time (Bruun and Hukkinen 2003). Although Bijker (1997, 126) attempted to articulate the content of interactions among social groups and the power structure under this process through the concept of "technological frame," this notion has been defined too broadly as "not purely cognitive, but also comprising social and material elements." However, this conceptual breadth has drawn criticism for its complexity (Prell 2009), particularly its conflation of cognition and practice.[2] We contend that this conflation has problematized the operationalization of some SCOT concepts in empirical research.

To address these limitations, this study introduces a more analytically tractable concept: imaginaries, while keeping SCOT's foundational commitment to treating technologies symmetrically as sociotechnical artifacts. Rather than examining the combined cognitive, social, and material elements that shape meanings and behaviors, we focus specifically on the cognitive dimension. In doing so, we investigate how programmers imagine what AI is and how it should or will affect their work. From this perspective, the concept of imaginaries provides a sharper and more suitable analytical lens than technological frames, especially on its isolating the future-oriented, cognitive dimension of collective perceptions on AI.

Originally associated with the meaning of "a mere semblance, unreal, fictitious, pretended," (McNeil et al. 2017, 436) the term "imaginaries" has gradually evolved into a conceptual tool for analyzing the human capacity to envision alternative futures. Within STS, imaginaries have become one of the most actively used concepts for empirical inquiry, due to its ability to articulate abstract yet socially consequential processes. Scholars have since developed a wide range of derivative notions—from scientific imaginary (Fortun and Fortun 2005; Wenzel Geissler and Prince 2009),

sociotechnical imaginaries (Jasanoff and Kim 2009), to AI imaginaries (Zhong et al. 2025)—to capture different dimensions and levels of meaning-making. However, the modifiers are often seen as either contextual, tied to specific empirical domains, or analytically redundant (McNeil et al. 2017). Against this backdrop, this study returns to the broader notion of imaginaries to explore how programmers envision emerging technologies like generative AI.

Building on this line of work, we apply the lens of imaginaries to enhance conceptual precision, particularly by isolating the future-oriented, cognitive dimension of how programmers engage with generative AI in their everyday practice.

AI imaginaries in China: national, public and professional perspectives

The final section reviews existing research on AI imaginaries in China across three interconnected levels: national, public, and professional. Given AI's rapid development and wide-ranging applications, studies at all three levels have increasingly centered on generative AI systems such as ChatGPT and DeepSeek, which now serve as the primary reference point across these domains.

At the national level, AI imaginaries in China mainly follow two trajectories. The first is a developmentalist vision, domestically oriented and framed around the state's ambition to harness AI as a morally guided engine of social and economic progress. This trajectory merges innovation with governance, reinforcing the state's central role in directing sociotechnical change (Bareis and Katzenbach 2022; Mao and Shi-Kupfer 2023; Zhu and Lu 2024). The second trajectory is outward-facing and geopolitical, portraying AI as a key instrument for national rejuvenation and technological sovereignty amid intensifying global competition, particularly with respect to China's positioning in the international AI landscape (McInerney 2024; Richter, Katzenbach, and Zeng 2025).

Public imaginaries of AI, in contrast, are grounded more firmly in daily encounters and concerns. These include expectations of efficiency and convenience, as well as anxieties about privacy, misinformation, and the diminishing distinctiveness of human cognition (Lian et al. 2024; Xu and Zhang 2026; Zou and Liu 2024). Such public perceptions are shaped by engagement with digital platforms and are further mediated by competing narratives from government agencies, academic institutions, commercial enterprises, and mass media (Meng, Zhang, and Cao 2024). Building on these mediated public discourses, Zhong et al. (2025) introduced the concept of "AI imaginaries" and proposed the "AI Imaginary Model" (AIM) to conceptualize how collective visions of AI emerge

through everyday sociotechnical experiences and, in turn, shape public users' technological identity, expectations, and cognitive engagement with AI systems.

In contrast to the rich body of research on national and public imaginaries, those of professional groups remain underexamined, even though such groups are critical in shaping the development and application of generative AI. Existing studies on professional AI imaginaries have concentrated mainly on journalists and educators, occupations where AI's integration is already visible and contested. Journalists present mixed feelings: some emphasize the efficiency gains AI brings to daily reporting, while others highlight ethical and legal risks (Kuai 2025). Educators, by contrast, exhibit more pronounced anxiety, reflecting concerns about both their own capacity to use AI tools effectively and the possibility that students might misuse these technologies or become overly dependent on them (Kohnke, Moorhouse, and Zou 2023).

Together, these studies suggest that professional imaginaries are shaped not only by the technical affordances of AI (Zhan et al. 2024), but also by occupation-specific values, skills, and responsibilities, an insight that is insufficiently addressed in the case of programmers, who are even more crucial than journalists and educators in influencing how AI evolves and is embedded in society.

Despite their pivotal position, scholars have paid little attention to the subjective domain of programmers' work and professional life, let alone their imaginaries of generative AI. Existing studies mainly address the objective dimensions of their work and status: one strand examines the labor process, such as modularized task division (Jalote and Kamma 2021; Yan and Wang 2025) and flexible working hour system (Liang 2019; Li and Liu 2020); the other focuses on occupational stratification, often along gender lines (Sun 2021; Wu 2020).

The few existing accounts of programmers' subjective orientations tend to be broad occupational culture overviews, outlining shared values, norms, and ethics within the profession (Li 2023a; Wu 2020). Programmers are often portrayed as exhibiting a strong faith in technological progress and a fascination with novel systems (Wang et al. 2024, 139–140). This technocentric orientation, combined with the demands of high-intensity work, drives continuous self-learning, both to keep pace with technological change and to maintain competitiveness (Wang and Yang 2021; Yan 2021). Within this ethos, self-improvement becomes not merely an external requirement but an internalized moral imperative (Luhr 2024).

Researchers also focused on the stratification of programmers: unlike many occupations historically stratified through bureaucratic hierarchies, programmers are primarily differentiated by technical expertise (Burrell and Fourcade 2021), resting less on formal titles than on the

accumulation of specialized knowledge, mastery of emerging tools, and proven problem-solving ability (Campero 2021). Higher proficiency and capacity open access to complex, high-impact projects, enabling employees to shape technical standards, mentor peers, and influence both the professional community and the broader tech industry.

Yet this very dependence on cutting-edge expertise places them at the forefront of technological upheaval: programmers are among the occupational groups (Zhang et al. 2025) and industries (Felten, Raj, and Seamans 2021) most directly affected by the disruptive potential of generative AI, as automation begins to reshape their forms of labor (Shestakofsky 2017). These dynamics make programmers not only key intermediaries in the development of generative AI, but also analytically crucial actors in understanding how AI is imagined, categorized, and socially embedded in practice.

In sum, while studies of AI imaginaries have increasingly incorporated professional perspectives, they have overlooked programmers, a structurally significant yet heterogeneous social group. This study addresses that gap by specifically focusing on programmers, analyzing how they imagine generative AI and how such imaginaries vary across different dimensions of technical expertise, including formal training, professional experience, and occupational role.

Research design

This study adopts a two-stage mixed-methods design to examine both the widespread imaginaries of AI among programmers and their distribution across technical strata within the programming profession (Figure 1). In the first stage, we descriptively explore a large corpus from a specialized programmer forum to identify major patterns of interpretation and discourse. In the second, we estimate the prevalence of these imaginaries across technical subgroups using a nationwide survey. This sequential design aligns discovery-oriented insights from digital trace data with

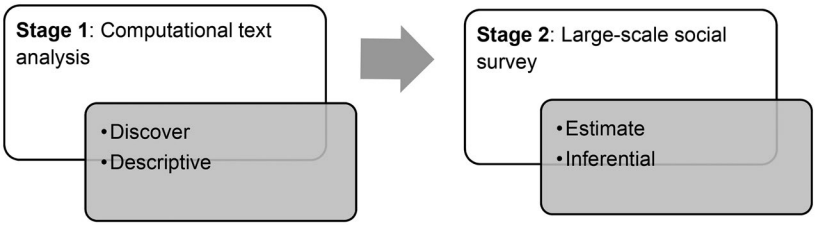

Figure 1. Design for the two-stage mix-methods study. This framework integrates large-scale textual and survey data to explore how Chinese programmers imagine and interpret generative AI.

statistically grounded survey evidence (Housley et al. 2014), jointly mapping the contours of programmers' AI imaginaries.

Computational text analysis of forum data

Our corpus is drawn from a publicly available repository of "Way to Explore" (V2EX), a leading Chinese online forum primarily used by professional programmers. Originally introduced as a platform for "designers, programmers, and creative individuals"[3] in 2005, V2EX has evolved into a space where user activity is overwhelmingly focused on software development.

As of 22 July 2023, when the dataset was extracted, V2EX comprised 194,534 registered users, 801,038 posts, and 10,899,382 replies, with the five most active categories being "develop," "app," "python," "mac," and "Java" (Oldshensheep 2023). Compared with other mainstream platforms for programmers, such as Stack Overflow, GitHub and CSDN, V2EX is characterized by a high concentration of discussion-oriented posts. On the other hand, unlike other social media platforms such as Weibo or RedNote (*Xiaohongshu*), its user base is more specialized and explicitly oriented toward technical dialogue. This combination of thematic focus, active engagement, and public accessibility makes V2EX a highly suitable site for examining how Chinese programmers perceive and interpret generative AI.

The target posts were identified through a keyword list derived from both Chinese and international studies on AI perceptions (see Figure 2

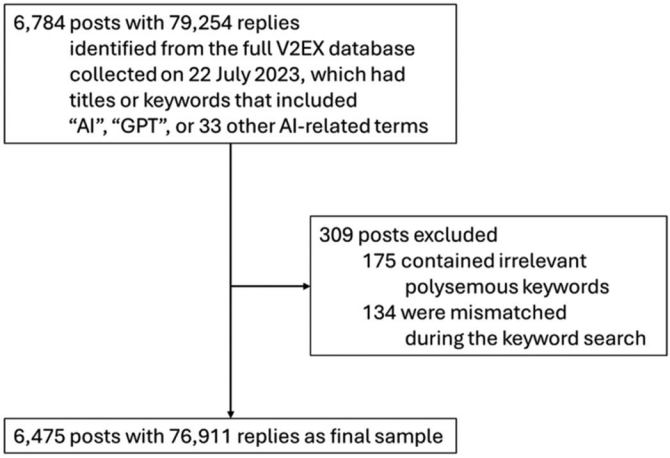

Figure 2. Flow diagram of the post selection process. This stepwise filtering ensured thematic relevance and high engagement, resulting in a robust corpus for text analysis.

for the post selection process and Table A1 in the Appendix for the selected keywords). An initial set of 6,784 posts was retrieved from the full dataset using this keyword filter. Subsequently, they were manually screened to exclude entries unrelated to artificial intelligence (for example, posts where "AI" referred to "Adobe Illustrator"). The final sample comprised 6,475 posts, totaling approximately 4.13 million words, including Chinese and English characters and punctuation.

Drawing on established computational approaches in sociology (DiMaggio, Nag, and Blei 2013; Pardo-Guerra 2020), this study employs an integrated computational-qualitative sequence to identify and interpret programmers' imaginaries of AI.

First, we applied an unsupervised machine learning topic model to the forum posts to inductively uncover latent thematic clusters. The topic modeling was performed using BERTopic, an algorithm that integrates clustering techniques with pretrained large language models (Grootendorst 2022). Unlike traditional methods such as Latent Dirichlet Allocation (LDA), BERTopic is well-suited to short, informal social media texts, as it captures semantic similarity beyond surface-level word co-occurrence (Egger and Yu 2022). Owing to these advantages, it has been increasingly adopted in recent computational social science research (Baribi-Bartov, Swire-Thompson, and Grinberg 2024; Garbes and Marlow 2025; Hou and Huang 2025), providing an effective means of extracting meaningful thematic patterns from large-scale online discourse.

Second, building on the clusters identified through BERTopic, we further conducted close readings of typical and influential posts, as indicated by engagement metrics, to situate these computationally derived clusters within their broader sociotechnical contexts, thus helping refine thematic interpretations and contextualize them within lived experiences of programmers.

The above analysis is based solely on publicly available online content, conducted in accordance with standard ethical protocols, and approved by the Science and Technology Ethics Committee (Humanities, Social Sciences and Engineering), Tsinghua University (Approval No. THU-04-2025-1000).

National social survey of Chinese programmers

To examine the distribution of the imaginaries identified through the forum-based computational text analysis across technical strata, the second stage is based on data from a large-scale online survey of Chinese programmers, collected between July and August 2023. The survey was designed and carried out by the Center for Social Survey and Research and the Department of Sociology, School of Social Sciences, Tsinghua

University, within a national research initiative on science and technology workforce development.

The questionnaire focused on the working conditions, skill structures, and value orientations of programmers, with a particular interest in assessing the impact of generative AI on programmers' labor processes. Accordingly, items were designed to probe this issue directly. Furthermore, the questionnaire included multiple indicators to capture the internal stratification of the programmer community, enabling us to measure their technical expertise from several distinct yet interconnected dimensions.

Given the uneven geographic distribution of China's software industry, particularly concentrated in the eastern region, the survey adopted a purposive sampling design with a quota-based distribution strategy to intentionally incorporate regional variation (Department of Operational Monitoring and Coordination 2024).[4] With support from a nationwide science and technology association, the questionnaire was distributed across key regions to reflect the structural layout of China's software sector.

After excluding responses completed by automation tools, partially filled questionnaires, respondents working in non-technical occupations, as well as regions with insufficient sample sizes (Gansu, Qinghai, Yunnan, Hainan, Inner Mongolia, and Tibet, each with fewer than 30 cases), the final analytic sample comprised 14,445 valid cases across 25 provincial-level administrative regions in the Chinese mainland. As one of the few large-scale social surveys to focus specifically on programmers, both in China and globally, this dataset offers a rare empirical vantage point for understanding the occupational realities of this key professional group within the ongoing digital transformation.

Ordinary Least Square (OLS) models were applied to analyze the dominant imaginaries among different groups. To achieve this, we translate the metaphorical distinctions observed in online discourse into a survey-based indicator that captures respondents' relative orientation toward the two imaginaries. Given that online discussions predominantly frame AI in instrumental terms, particularly in relation to programming tasks, we measured programmers' attitudes toward AI using the following survey item: "To what extent do you think generative AI (e.g., ChatGPT) poses a threat to your profession?" Responses were recorded on a five-point Likert scale ranging from "Much more of a threat than a help" (1) to "Much more of a help than a threat" (5), with higher scores reflecting a stronger belief that AI is a help.

The key independent variable is technical expertise, which, as discussed earlier, constitutes a core axis of internal stratification among programmers (Burrell and Fourcade 2021). We operationalize technical expertise through three binary indicators that capture complementary dimensions of technical capacity: first, educational background:

whether the respondent holds a degree in computer science or software engineering (1 for yes, 0 for no); second, job type of programming: whether the respondent occupies an advanced technical role—defined as system design, artificial intelligence, or algorithm development (1 for advanced roles, 0 for other positions, including front-end and back-end development, visual design, data work, operations and maintenance, and software testing); third, years of professionally programming: whether the respondent has more than four years of programming experience, measured from the start of their formal career, which corresponds to the median tenure among all surveyed programmers (1 for four years or longer, 0 for less than four years). Each indicator takes the value of one to denote a higher level of technical expertise. Together, they represent layered dimensions of technical expertise: formal education, functional responsibility, and accumulated experience.

In addition to the key independent variables, we further control for a broad set of sociodemographic characteristics, comprising gender, age, type of *hukou*, marital status, number of children, and years of education, as well as work-related attributes, encompassing type of employer, monthly income, professional rank, weekly workhours, number of projects, and number of team members, as these variables are considered influential for occupational outcomes among programmers (Yan and Wang 2025). Descriptive statistics for all variables are reported in Table B1 of the Appendix.

Results and findings

Overview of discussions and topic modeling

To provide an initial overview, we present the quarterly volume of AI-related posts on V2EX, juxtaposed with the Baidu Search Index on AI, rescaled to match the range of V2EX posts, to facilitate comparison with public search interest. Figure 3 reveals three waves of attention: an initial rise in 2016 (coinciding with AlphaGo's match against Lee Sedol), a secondary peak in 2019 (around Baidu's AI conference), and a dramatic surge in late 2022 with the release of ChatGPT.

From these temporal patterns, two features are particularly noteworthy. First, the 2023 wave dwarfs earlier activity—posts jump from a local peak of 168 in the third quarter of 2019 to 1,879 in the second quarter of 2023 (over tenfold). Second, it is only with the advent of generative AI that programmer engagement with AI becomes both massive and sustained. Prior to the end of 2022, AI was discussed intermittently and in modest volumes, even after high-profile events. The post-ChatGPT surge thus

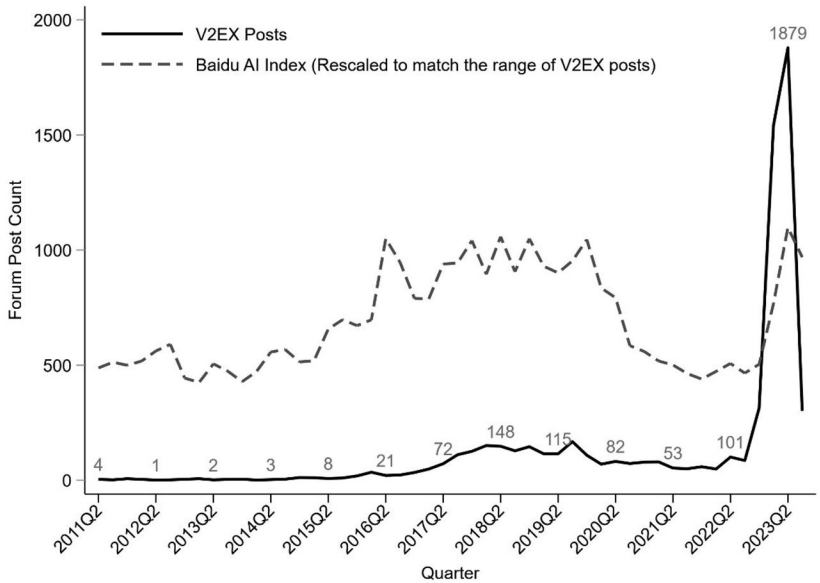

Figure 3. Temporal dynamics of V2EX posting and Baidu search index for "AI". Both show sharp increases around late 2022, indicating a surge in public and professional interest following the release of ChatGPT.

marks a qualitative shift: from episodic curiosity about AI in general to intensified and ongoing reflection on its technical potential, limitations, and implications for their own work.

Building on the above overview, we now turn to the substantive content of their discussions. We applied BERTopic to the corpus, which generated five coherent clusters, two of which contain several sub-clusters that map the core themes of programmers' AI-related discourse (Table 1). The thematic labels were manually developed by the authors. As Lee and Martin (2015) note, topic models do not uncover semantic meaning but identify clusters of co-occurring terms that suggest latent patterns of shared interpretation.

The largest topic cluster, accounting for nearly half of the posts (3,043 out of 6,475), focuses on AI research and development (R&D) and programmer employment. Its most representative terms include "(software) development," "technology," and "familiarity." Within this cluster, two employment-oriented sub-clusters (T1-1 and T1-2, with 123 and 264 posts respectively) deal with skill requirements and experience thresholds for AI employees. In these sub-clusters, discussions frame AI primarily through its labor-market relevance—portraying it as a technical asset, a skill to acquire, or a tool to master—hence a distinctly instrumental orientation.

Table 1. BERTopic-clustered AI-related discussions on V2EX.[a]

Topic hierarchy	Thematic label	Number of posts (N)	Top 10 representative terms
Main Topic 1	AI R&D and Programmer Employment	3,043	AI; (software) development; technology; artificial intelligence; familiarity; work; algorithm; product; related; capability.
Sub-topic 1-1	Programmer Employment	123	Experience; (software) development; familiarity; capability; priority; work; related; product; technology; algorithm.
Sub-topic 1-2	Programmer Demand	264	Engineer; experience; company; work; algorithm; technology; familiarity; development; recruitment; team.
Sub-topic 1-3	Societal Impact & Future Development	211	Human being; artificial intelligence; AI; possibility; without; now; (future) development; should; robot; future.
Main Topic 2	Ecosystem & Practices of ChatGPT	3,071	ChatGPT; usage; API; support; issue; need; model; everyone; without; code.
Sub-topic 2-1	API Access & Model Versions (GPT-4)	341	GPT-4; GPT; API; 4.0; quota; usage; account; Plus; application; issue.
Sub-topic 2-2	Auxiliary Functions & Learning Tools	290	ChatGPT; plugin; code; learning; usage; issue; English; translation; everyone; (software) development.
Sub-topic 2-3	Account Registration & Payment Channels	159	Plus; credit card; DePay; ChatGPT Plus; virtual; top-up; payment; ChatGPT; activation; binding.
Main Topic 3	Software Usage & Troubleshooting	178	Code; usage; without; discover; briefly; feeling; issue; open source; login; today.
Main Topic 4	Web Development & Remote Work Opportunities	10	JavaScript; technology; company; remote; maintenance; product; team; possess; work; data.
Main Topic 5	AI Hardware and I/O Configuration	39	System; AI; connection; output; input; two; issue; this type; copy; correct.
Unclustered Topic	N/A	134	N/A

[a]Includes replies before the captured date, 23rd July, 2023; Original Chinese words in Table 2: (software) development(开发), technology(技术), artificial intelligence(人工智能), familiarity(熟悉), work(工作), algorithm(算法), product(产品), related(相关), capability(能力), experience(经验), priority(优先), engineer(工程师), company(公司), recruitment(招聘), team(团队), human being(人类), possible(可能), without(没有), now(现在), (future) development (发展), should(应该), robot(机器人), future(未来), usage(使用), support(支持), issue(问题), need(需要), model(模型), everyone(大家), code(代码), quota(额度), account(账号), application(申请), plugin(插件), learning(学习), English(英语), translation(翻译), credit card(信用卡), virtual(虚拟), top-up(充值), payment(支付), activation(开通), binding(绑定), discover(发现), briefly(一下), feeling(感觉), open source(开源), login(登录), today(今天), remote(远程), maintenance(维护), possess(具有), data(数据), system(系统), connection(连接), output(输出), input(输入), two(两个), this type(这种), copy(复制), correct(正确). Both "AI" and "artificial intelligence" were retained in the topic modeling process to preserve the integrity of the original data and were treated as semantically equivalent during interpretation.

Another notable sub-cluster (T1-3, 211 posts) includes keywords such as "human being," "robot," "possibility," and "future," suggesting broader imaginaries concerning AI's evolving role. In addition to the labor-market focus, this cluster indicates philosophical contemplations on AI's societal position and its relation to human existence. Taken together, main topic 1 (T1) reflects a dual orientation: instrumental applications co-exist with speculative or philosophical reflection.

Main topic 2 (T2) centered on the ChatGPT application ecosystem, comprises 3,071 posts and consists of three sub-clusters: API and model version updates (T2-1, 341 posts), auxiliary functions and learning tools (T2-2, 290 posts), and account or payment issues (T2-3, 159 posts). Collectively, these sub-clusters highlight the forum's continued pragmatic ethos, as users actively co-construct informal usage manuals. Representative terms such as "API," "quota," "plugin," "translation," and "payment" indicate a strong emphasis on technical onboarding, productivity enhancement, and access coordination.

The remaining clusters address more specific technical challenges. These include main topic 3 (T3), software troubleshooting (178 posts), main topic 4 (T4), web development and remote work opportunities (10 posts), and main topic 5 (T5), AI hardware and I/O configuration (39 posts). Keywords such as "code," "remote," and "connection" further underscore the pragmatic orientation that characterizes programmers' engagement with generative AI.

Overall, across the five topic clusters, programmers' imaginaries of generative AI indicate three defining aspects: an instrumentalist, pragmatic-oriented framing; a labor-centered, recruitment-related context; and reflexive engagement.

The first of these, instrumentalism combined with a pragmatic orientation, frames generative AI foremost as a means of problem-solving. Consistently imagined as a learnable and integrable tool, AI is embedded in daily technical routines, from specific applications (T2) and configuration (T5) to troubleshooting (T3), as reflected in the recurrence of technical keywords such as "(software) development," "plugin," and "API" across multiple clusters.

The second foregrounds AI's entanglement with the labor process. For programmers, employment emerges as their most immediate concern, reflected in recruitment-related topics (T1-1, T1-2, T4), signaling AI's growing presence in job-seeking and hiring dynamics.

The third feature is reflexive engagement. While largely bounded within the technical labor process, these imaginaries are not devoid of reflexivity. Posts in T1-3 concerns about AI's implications for human cognition and reflect on its ethical and ontological dimensions. Such moments of speculation, though intermittent and interwoven with an instrumentalist outlook, allow Chinese programmers to move beyond task-specific concerns and articulate early positions on AI's societal implications, from its effects on human cognition to its broader normative and philosophical ramifications.

Internal tensions: anxiety under instrumentalism

While topic modeling delineates the structural contours of major discussion areas, it cannot, on its own, uncover the affective substance and discursive logics that animate them (DiMaggio, Nag, and Blei 2013). To

Table 2. Synopsis and key excerpts from the two representative posts (translated).[a]

	Post #900126 (100,947 clicks; 374 replies)	Post #900396 (22,842 clicks; 149 replies)
Title	OpenAI releases groundbreaking ChatGPT: A registration guide	Experiencing ChatGPT: The more I played, the more I have existential concerns
Post date	December 5, 2022	December 6, 2022
Main Content	A few days ago, OpenAI released the much-anticipated ChatGPT, which quickly became very popular. However, due to network restrictions, most users could not access it directly. Below is a concise guide to registering for ChatGPT (*details omitted*): – Prerequisites – Obtain a temporary foreign number – Sign up for an OpenAI account – Phone verification – Start using ChatGPT	AI is advancing even faster than I imagined – I don't know what we'll do in the future. At first I was genuinely amazed by what ChatGPT could do, though many commenters found it underwhelming – I suspect they simply didn't ask the right questions. Its open-ended conversational style is nothing like the old Q&A bots that just parsed language and hooked into a search engine. In any case, I believe the "Spinning Jenny" moment is here – or about to arrive. Artisans may lose their jobs, but countless operator roles will be created. We shouldn't be pessimistic about rising productivity; we should be excited.
Representative replies	Reply #900126-1: "You lot are going to kill yourselves sooner or later" Reply #900126-11: "Thanks, it has been done" Reply #900126-25: "I'm already using this to help me write code 😊" Reply #900126-51: "I'm already using it—even got the bot to draft my year-end summary…" Reply #900126-166: "Regex lookup? Fake review generation? Betting predictions? Doesn't feel very meaningful… over 100 replies—what's all the fuss?"	Reply #900396-1: "Programmers are definitely going to be replaced" Reply #900396-9: "Most of the replies just go round in circles—verbose but unfocused, and less substantive than even a manager's meeting. What is there to worry about?" Reply #900396-11: "The more creative the work, the harder it is for AI, because it only operates on the 'corpus' it's been fed—finite and known—whereas creativity is infinite" Reply #900396-18: "If we define innovation as 'the recombination of old (existing) elements,' do you still say AI can't innovate?" Reply #900396-34: "LMAO. Would you race a car to see who's faster? Tech shifts bring change, but our only real rivals are other humans."

[a]Includes only replies before the captured date, 23 July, 2023.

address this limitation, we turn to a close reading of two among the most-viewed AI-related posts on V2EX (Table 2). This micro-level analysis reveals a striking undercurrent: even within a largely instrumentalist and pragmatic community, the most widely engaged discussions are suffused with unease, anxiety, and moments of critical reflection, collectively forming a latent tension that underlies programmers' imaginaries of generative AI.

Post #900126, titled "OpenAI Releases Groundbreaking ChatGPT: A Registration Guide" (5 December 2022), drew 100,947 views and 374 replies within days of ChatGPT's release. Initially a step-by-step guide to circumvent access restrictions, the thread quickly turned into a forum for divergent reactions to generative AI. Some replies struck an alarmist tone, offering visceral warnings rather than analytic critique (Reply #1). Others

endorsed ChatGPT as a practical aid for coding and work-related writing (Replies #25, #51). A skeptical strand also appeared, downplaying the tool by pointing to mundane uses such as regex lookups or fake-review generation (Reply #166). Overall, the exchange reveals growing ambivalence: programmers oscillate between anxiety, enthusiasm, and irony as they weigh ChatGPT's implications for everyday coding practices and the broader trajectory of technological development.

On the following day, Post #900396, titled "Experiencing ChatGPT: The More I Played, the More I Have Existential Concerns," received 22,842 views and 149 replies. The author described a "Spinning Jenny moment," likening ChatGPT's emergence to the mechanization of textile labor during the Industrial Revolution. While acknowledging the risk of job losses, the post remained optimistic, highlighting the potential productivity gains and the creation of new roles. The first reply, which stated that "programmers are definitely going to be replaced," nonetheless set a markedly pessimistic tone. Some respondents embraced this negative outlook (Reply #18), whereas others echoed the author's guarded optimism or offered more skeptical takes (Replies #9, #11, #34). These discussions, like those in the previous post, again reveal the coexistence of reassurance and anxiety, underscoring the broader uncertainty over how generative AI will reshape programmers' work.

Diversity of AI imaginaries: AI-as-machine vs. AI-as-intelligence

Drawing on these tensions, we conceptualize programmers' imaginaries using two metaphorical ideal types: "AI-as-Machine" and "AI-as-Intelligence." Together, these orientations capture both the labor-centered focus and the ambivalent emotional tone evident across the dataset. While both imaginaries treat AI as a tool, they differ in how they situate that tool in relation to human labor and cognitive authority. The "machine" imaginaries cast AI as a sophisticated yet ultimately manageable extension of automation. By contrast, the "intelligence" imaginaries embody anxieties that the tool may already possess forms of autonomous reasoning that threaten the very foundations of specialized and creative work.

These metaphors are not merely rhetorical; they are rooted in core concepts from STS, where imaginaries of "machine" and "intelligence" reflect deeper assumptions about control, agency, and human–technology relations. The "machine" metaphor, in particular, emerges from an industrial-era imaginary of technology as neutral, passive, and governable. Consistent with Mumford's (1934, 9–12) view of the machine as a device for executing fixed routines, this imaginaries center on repetition, predictability, and human command. As Feenberg (1999, vii) also argues,

modern societies have long linked machines to instrumental rationality—privileging efficiency, control, and standardization over ambiguity and reflexivity. Within this lineage, the AI-as-Machine imaginaries reduce AI to a rationalized, predictable apparatus, mirroring the bureaucratic logics of factory discipline and administrative control. Even as generative AI deviates technically from these mechanical systems, programmers may still interpret it through the inherited lens of machinic subordination.

The "intelligence" metaphor, in contrast, emerges from imaginaries of technology as adaptive, self-directed, and increasingly autonomous. Rooted in the idea that modern technologies may acquire the appearance of autonomous systems (Winner 1978), these imaginaries frame generative AI as a system that evolves alongside its users.

While generative AI systems, and large language models (LLMs) in particular, remain fundamentally algorithmic in operation, governed by probabilistic models, recursive training cycles, and parameterized inference mechanisms, they depart from earlier back-end algorithms in visibility and interactivity. Unlike conventional systems that operated largely behind the scenes, generative AI tools directly engage users, exhibit self-refining behaviors, and appear to "learn" through interaction. This interactive quality intensifies perceptions of autonomy and co-evolution, lending credence to the "AI-as-Intelligence" imaginaries (Schinkel 2023).

Critical inquiry therefore often targets these algorithmic substrates rather than the branded artifacts marketed as "artificial intelligence" (Burrell and Fourcade 2021; Schinkel 2023). Within this discourse, generative AI can be seen as a new kind of "autonomous technology," not only in terms of its cognitive affordances but also in the way it is discursively framed as a self-perpetuating system beyond deliberate human control. Yet as Wachter-Boettcher (2018, 96) warns, such perceptions risk obscuring the biases and institutional power embedded in algorithmic design, reinforcing the illusion of technical objectivity and inevitability.

We argue that "machine" and "intelligence" encapsulate Chinese programmers' imaginaries because they are emblematic metaphors of two distinct technological eras, each imbued with layered attitudes and emotions. The five metaphorical pairs in Table 3 are not merely descriptive contrasts but constitute a schema through which programmers make sense of AI. Spanning behavioral, functional, epistemic, developmental,

Table 3. Metaphoric contrasts between "machine" and "intelligence."

Dimension	Machine	Intelligence
Behavioral	Passive	Active
Functional	Executive	Interpretive
Epistemic	Interpretable	Uninterpretable
Developmental	Degrading	Evolving
Political	Controllable	Uncontrollable

and political dimensions, these pairs chart an imagined shift from the industrial-era logic of the "machine" to the digital-era vision of "intelligence."

These two metaphors also reflect fundamentally different relational stances toward generative AI, inductively distilled from programmers' forum discussions. The "machine" metaphor connotes repetition, predictability, and subordination, where tools that remain firmly under human control. While digital infrastructures such as cloud computing and storage, model-as-a-service (MaaS) may improve over time through scaling and recursive optimization, our focus is not on material wear but on the symbolic qualities of passivity and manipulability that programmers attach to generative AI. By contrast, the "intelligence" metaphor signals adaptability and autonomy, evoking systems that co-evolve with users and resist human mastery. Notably, however, some programmers describe AI agents as intelligent or autonomous while expressing little sense of threat or displacement. In such cases, we still categorize these as "AI-as-Machine" imaginaries, since the underlying orientation remains tool-like and instrumental, maintaining human primacy over sociotechnical control. Taken together, these metaphors do more than describe technical functions—they structure how programmers imagine AI's development, governance, and implications for labor.

In sum, many Chinese programmers approach AI instrumentally, viewing it as a machine that offloads routine tasks, integrates new tools into existing workflows, enhances productivity, and supports skill acquisition. Yet beneath this pragmatic stance lies an alternative discourse that frames AI as genuine "intelligence," provoking anxieties over job security, professional identity, and the epistemic trustworthiness of AI-generated content. These diverging imaginaries form the conceptual scaffold for the next stage of our analysis, which examines their distribution across programmers with varying levels of technical expertise.

AI-as-machine: the dominant imaginaries among Chinese programmers

Building on the preceding computational text analysis, which inductively identified two imaginaries of generative AI among Chinese programmers, "AI-as-Machine" and "AI-as-Intelligence," we now turn to a large-scale survey to assess their prevalence. This quantitative analysis allows us to move from a qualitative mapping of metaphors to an operationalized measure, enabling the examination of how these imaginaries are distributed across programmers with varying levels of technical expertise.

We begin by outlining the basic characteristics of the surveyed programmers. Consistent with existing research, over 70% of respondents

were male, with an average age around 35. Most held at least a bachelor's degree, nearly half reported a monthly income exceeding 10,000 RMB, and close to 60% worked more than 40 hours per week.

Regarding the key explanatory variables capturing professional expertise, 59.11% of respondents held a degree in computer science or software-related fields, while 40.89% came from other disciplines. In terms of job type, 10.81% were employed in advanced technical positions, with the remaining 89.19% occupying other roles. As for professional experience, 54.43% had worked in the industry for four years or more, whereas 45.57% had less than four years of experience. Descriptive statistics for all variables are presented in Table B1 of the Appendix.

When it comes to their attitudes toward generative AI, on average, respondents scored 3.16 on the measure of their technical expertise ($SD = 1.02$), suggesting a moderately positive orientation toward generative AI. Approximately 38.08% of respondents rated AI as more helpful than threatening (scores 4 ~ 5), 42.06% expressed a neutral stance (score 3), and 19.86% perceived it as more threatening than helpful (scores 1 ~ 2).

In this study, "AI-as-Machine" imaginaries are operationalized as perceiving generative AI to be more helpful than threatening, whereas "AI-as-Intelligence" imaginaries are operationalized as perceiving it to be more threatening than helpful. The "machine" logic captures a "practice makes perfect" orientation, in which generative AI is viewed as a tool that can be mastered and kept subordinate to human control. Individuals who regard generative AI as more helpful than threatening are therefore understood to hold these imaginaries. By contrast, the "intelligence" logic envisions AI as an evolving and self-improving entity whose users effectively act as "trainers," thereby heightening concerns that it may eventually challenge human authority or expertise, resonating with the adage that "ignorance breeds fearlessness." In this operationalization, higher scores correspond to the "AI-as-Machine" imaginaries, whereas lower scores align with the "AI-as-Intelligence" imaginaries.

Stratified imaginaries: variations across different levels of expertise

OLS regression was employed to examine the distribution of different imaginaries. Before introducing the key independent variables, we first estimated a null model (model 0) containing only the control variables. Figure 4 visualized the results of this model, illustrating the relationships between various socioeconomic characteristics, working conditions, and attitudes toward generative AI. Except for professional rank, all control variables show significant associations with the dependent variable, underscoring the necessity of including them in the model.

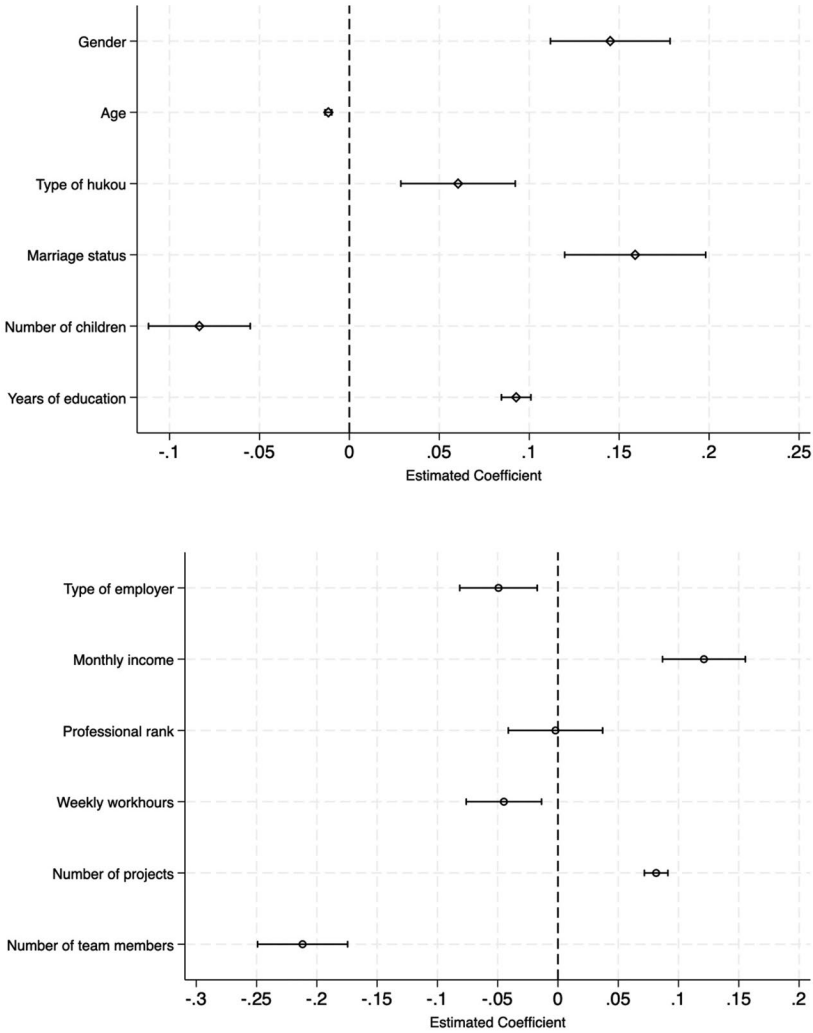

Figure 4. Attitudes toward generative AI among Chinese programmers with different socioeconomic statuses (top) and work-related attributes (bottom). Higher education, income, and managerial status are associated with more positive orientations toward AI, while heavy workloads and lower professional ranks correlate with greater negative orientations. *Note:* Estimated coefficients and 95% confidence intervals are derived from regression models (see Table C1 in the Appendix for full results).

Building on model 0, model 1 adds three key independent variables. As shown in Figure 5, the results consistently suggest that programmers in structurally advantaged technical positions are more inclined to perceive generative AI as helpful. Those with a computer science or software--related educational background report significantly higher levels of perceived help ($\beta = 0.106$, $p < 0.001$). Likewise, individuals in advanced

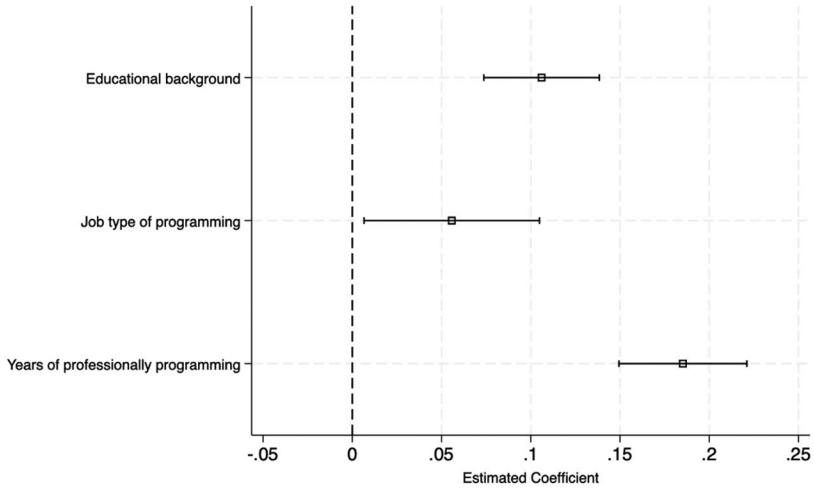

Figure 5. Attitudes toward generative AI among Chinese programmers with different levels of technical expertise. Programmers with advanced training, formal CS degrees, and longer experience tend to express more favorable perceptions of AI's utility and less anxiety over its threats. *Note:* Estimated coefficients and 95% confidence intervals are derived from regression models (see Table C1 in the Appendix for full results).

technical roles express greater optimism than their counterparts in other job categories ($\beta = 0.056$, $p = 0.026$). Programmers with at least four years of professional experience also report markedly higher perceived help than those with less experience ($\beta = 0.185$, $p < 0.001$). Collectively, these findings reflect that programmers with stronger technical expertise and greater occupational seniority are more confident in engaging with generative AI and are therefore more likely to adopt the "AI-as-a-Machine" imaginaries.

This orientation aligns with a traditional instrumentalist understanding of AI: as an augmentative force in human labor rather than an autonomous, creative intelligence. In this framing, rooted in industrial-era technological imaginaries, machines are conceived as extensions of human capability rather than independent agents. Such a view stands in contrast to the "AI-as-Intelligence" imaginaries.

Notably, technically proficient programmers, positioned as key actors within digital production systems, tend to hold markedly more optimistic views. From a SCOT perspective, such optimism is not merely an individual disposition but a structuring force that is likely to shape the trajectory of this emerging technology. We return to this implication in the discussion.

Conclusion and discussion

The rapid proliferation of generative AI has sparked growing scholarly interest in how such technologies are imagined by the state and the

general public. Unlike earlier waves of revolutionary technology, such as nuclear power (Bodrunova 2012; Fang 2014), genetically modified organisms (Tao and Shudong 2003; Guo 2005), and electric vehicles (Anyadike-Danes 2024; Xing 2024), which were largely confined to specialized domains with clear points of regulation, generative AI is characterized by its continuous, pervasive, and increasingly personalized integration into everyday life. This proximity to daily life not only reshapes public and professional encounters with technological change, but also heightens the urgency of examining AI's broader social implications.

As a key social group positioned between upstream innovation and downstream application, programmers wield considerable influence over how generative AI is socially embedded and operationalized. Yet the imaginaries of programmers remain underexplored. To address this gap, this study adopts a mixed-methods approach, combining computational topic modeling and close reading of forum posts with survey-based regression analysis, to investigate how Chinese programmers imagine generative AI.

Based on our empirical findings, we draw two central conclusions. First, through BERTopic modeling and close reading of influential posts, we identify two co-existing imaginaries among Chinese programmers: an instrumental framing of "AI-as-Machine" and a reflective framing of "AI-as-Intelligence." The former envisions AI as a passive, task-specific, and controllable tool oriented toward productivity enhancement, while the latter conceives of AI as an active, generalized, and self-evolving agent with the potential to challenge human cognition and subjectivity. These imaginaries are simultaneously articulated in programmers' discourse, reflecting both pragmatic engagement and deeper ontological concern.

Second, regression analysis shows that programmers with higher technical expertise are significantly less likely to view AI as a threat to their jobs. This pattern suggests that professional mastery fosters both professional confidence and psychological resilience, thereby reinforcing an instrumental "AI-as-Machine" imaginaries.

This pattern aligns with recent arguments that large AI models are not merely technical systems but cultural and social technologies whose meaning and usage are shaped by user knowledge and institutional positioning (Farrell et al. 2025). From a SCOT perspective, highly skilled programmers often operate as both users and producers of generative AI. Based on our national survey of Chinese programmers, we define such individuals as those working in AI development, algorithmic engineering, or architectural design. They train models, design architectures, and fine-tune applications. Their daily interactions with LLMs involve backend parameters and constraints (e.g., compute-data-parameter tradeoffs,

training loop optimizations, or alignment interventions), rather than abstract anthropomorphic traits. Such expertise anchors their perception of AI not as an autonomous intelligence, but as a technical instrument embedded in modular workflows. As a result, their imaginaries of AI are less mythologized and more materially grounded, reinforcing an instrumental, "AI-as-Machine" view.

In this sense, Chinese programmers' imaginaries of generative AI are not only interpretive but also stratified, shaped and distributed unevenly along lines of technical proficiency. This pattern offers a window into the emerging stratification within the programming profession. As highly skilled programmers increasingly view generative AI as an assistive tool that enhances their work, lower-skilled programmers who perceive generative AI as a threat may be more inclined to leave technical roles and redirect their coding skills toward adjacent or non-technical occupations. These divergent imaginaries, in turn, may interact with the material dynamics of automation, jointly reshaping the internal structure of the programmer community.

Our findings contribute to STS in both theoretical and empirical terms. Theoretically, this study links the SCOT framework with the more analytically tractable concept of imaginaries, thereby responding to recent critiques of SCOT (Basu 2023) as well as to Bijker's (2010) own efforts to integrate new concepts into the framework. Drawing on original survey data, we demonstrate that imaginaries among programmers are internally stratified according to levels of technical expertise. This highlights the social structuring of AI imaginaries and directly addresses one of SCOT's long-standing limitations, namely its neglect of stratification within relevant social groups. Finally, by focusing not on states or nations but on programmers as a strategically important professional community, our study answers recent calls to extend the study of imaginaries beyond abstract macro units to more concrete and agentive collectives (Kuchler and Stigson 2024).

Some scholars have cautioned that imaginaries risk appearing overly abstract if detached from material practices (Mager and Katzenbach 2021; Rahm and Rahm-Skågeby 2023). Munn (2020), for example, addresses this by developing the notion of operational imaginaries, showing how infrastructures concretely enact visions of the future in the present. Our study contributes to this effort by using concrete metaphors, namely, "machine" and "intelligence" (Table 3) to show how programmers' imaginaries of generative AI are similarly articulated and specified in their professional practices. Machines are predictable, passive, and tend to wear down over time. Mastery over machines follows an industrial logic of repetition and habituation: the more one engages with them, the more effective and precise one becomes in exerting control. In contrast, "intelligence" evokes a different logic: it implies an adaptive, learning entity

whose performance improves through interaction. Intelligent agents are not merely operated; they co-evolve with users and may develop a degree of autonomy and unpredictability that challenges traditional models of tool-use, raising questions about uncertainty and the erosion of human authority. This semantic contrast thus provides a meaningful lens through which to interpret the empirical predominance of the "AI-as-Machine" framing among Chinese programmers, making imaginaries not only conceptually relevant but also experientially grounded.

Empirically, our analysis of Chinese programmers offers distinctive insights with broader comparative relevance. Previous comparative studies of China's AI imaginaries have largely adopted a macro-level perspective, showing how the Chinese government and media portray AI as a positive and pragmatic driver of national development (Meng et al. 2024; Wang and Downey 2025). Our research complements this work by adding a micro-level dimension. As both creators and diffusers of AI technologies, programmers, especially those at senior technical levels, display a distinctive optimism and instrumentalism that may help explain the wider societal enthusiasm for AI in China, though this relationship warrants further exploration.

In summary, although this study captures a snapshot taken within roughly six months of ChatGPT's emergence, the imaginaries framework provides a conceptual anchor for projecting future trajectories. The relative absence of fear, particularly among technically proficient Chinese programmers, signals a distinctive cultural and professional disposition toward AI: one that treats it less as a disruptive threat and more as a malleable tool to be mastered. Tracing whether this composure endures, intensifies, or gives way to more ambivalent attitudes will be critical for understanding the long-term social embedding of generative AI.

At the same time, our analysis itself represents a snapshot—a node in the evolving history of generative AI imaginaries. As a concept oriented toward the future, imaginaries inevitably combine elements of prescience and naive. Much like revisiting Negroponte's (1995) *Being Digital* decades later reveals both perceptive insights and unfulfilled hopes, our aim is not to predict AI's future but to foreground how social meaning-making shapes technological pathways. In this sense, mapping Chinese programmers' current imaginaries also provides a window into the broader logics guiding China's AI development.

To advance a deeper understanding of generative AI beyond its technical features, future research should situate professional imaginaries within specific institutional contexts. Classical sociological theories of labor, from Marx and Braverman's critiques of deskilling to Thompson's (1963) emphasis on workplace resistance, have long viewed technology not as an isolated force but as one that is deeply rooted in social

relations. As generative AI becomes increasingly pervasive, scholars should investigate how it reconfigures professional hierarchies and redefines technical expertise. This study only serves as an entry point, constrained by its reliance on cross-sectional data. A further limitation of this study lies in its reliance on a single online forum (V2EX) for the first-stage analysis. While V2EX provides rich insights into programmers' discourse, it does not capture the full diversity of technical communities. Future research should therefore integrate additional platforms such as GitHub or CSDN to obtain a more comprehensive view.

Future research would benefit from updated data sources that enable longitudinal tracking of how professional imaginaries translate into concrete labor-market shifts as generative AI adoption deepens. Such data could illuminate whether lower-skilled programmers are increasingly exiting technical roles, being reassigned to adjacent tasks, or adapting through upskilling, thus empirically validating the stratification dynamics discussed in this study. In parallel, linking with digital trace data from platforms like GitHub or CSDN could reveal how imaginaries shape real-world programming practices. This integration would help unpack the recursive pathways through which expectations about generative AI not only influence individual behavior but also feed back into the technical development and organizational embedding of AI systems.

Notes

1. For convenience, this article employs "programmer" in a broad sense, as articulated by Li (2023b), to encompass related occupations such as software professionals, software engineers, and coders.
2. For instance, in information systems research, where the concept of the technological frame is widely applied, it is variably defined as either a purely cognitive construct or a hybrid of cognition and action (Nocera, Dunckley, and Sharp 2007).
3. See V2EX's introduction in Wikipedia: Wikipedia contributors. "V2EX." Last Modified July 2, 2025, at 03:46. https://zh.wikipedia.org/wiki/V2EX.
4. In this study, drawing on the current regional classifications used in official statistics and considering the actual spatial distribution of software-related enterprises, we collected data from four major regions, covering selected provincial-level administrative units within each, as follows:
   1 Eastern (9 provinces/municipalities): Beijing (municipalities), Tianjin (municipalities), Hebei, Shanghai (municipalities), Jiangsu, Zhejiang, Fujian, Shandong, Guangdong.
   2 Central (6 provinces): Shanxi, Anhui, Jiangxi, Henan, Hubei, Hunan.
   3 Western (7 provinces/autonomous regions/municipalities): Guangxi (autonomous regions), Chongqing (municipalities), Sichuan, Guizhou, Shaanxi, Ningxia (autonomous regions), Xinjiang (autonomous regions).

4 Northeast (3 provinces): Liaoning, Jilin, Heilongjiang.
Taiwan, Hong Kong and Macau are generally reported separately and not included in statistics on the Chinese mainland.

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Appendix

Post selection procedure

Table A1. Criteria for keyword selection.

Category	Keywords	Sources
General concepts	人工智能, Artificial intelligence, AI, 大语言模型, 大模型, large language model, LLM, LLMS, AI 生成内容, AIGC	Meng, Zhang, and Cao 2024; Zhang et al., 2024
Specific AI products	GPT, bing chat, bingchat, perplexity, 文心一言, ERNIE, 豆包, Doubao, 通义千问, qwen, 智谱清言, ChatGLM, 月之暗面, Kimi, bard, gemini, claude, dall, midjourney, imagen, craiyon, STABLE DIFFUSION, STABLEDIFFUSION, copilot, copilot	Lian et al. 2024; Miyazaki et al., 2024

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Descriptive statistics of key variables

Table B1. Descriptive statistics of key variables.

Variables		Frequency	Percent (mean)	*SD*	Min	Max
Attitudes toward generative AI		14,445	3.16	1.02	1	5
	Much more of a threat than a help	1,353	9.37%			
	More of a threat than a help	1,516	10.49%			
	Hard to say/Not sure	6,075	42.06%			
	More of a help than a threat	4,501	31.16%			
	Much more of a help than a threat	1,000	6.92%			
Educational background	Computer or software related	8,539	59.11%			
	Not computer or software related	5,906	40.89%			
Job type of programming	Advanced	1,562	10.81%			
	Other	12,883	89.19%			
Years of professionally programming	4 years or longer	7,862	54.43%			
	Less than 4 years	6,583	45.57%			
Gender	Female	4,130	28.59%			
	Male	10,315	71.41%			
Age		14,445	35.18	11.20	15	68
Type of *Hukou*	Urban	8,252	57.13%			
	Rural	6,193	42.87%			
Marriage status	Married at least once	7,272	50.34%			
	Never married	7,173	49.66%			
Number of children		14,445	0.69	0.78	0	3
Years of education		14,445	15.56	2.54	9	21
Type of employer	Public sector	5,867	40.62%			
	Private sector	8,578	59.38%			
Monthly income	10,000 *yuan* RMB and more	7,027	48.65%			
	Less than 10,000 *yuan* RMB	7,418	51.35%			
Professional rank	Higher	3,053	21.14%			
	Other	11,392	78.86%			
Weekly workhours	No more than 40 hours	6,092	42.17%			
	More than 40 hours	8,353	57.83%			
Number of projects		14,445	3.58	1.66	1	6
Number of team members	No more than 5	3,866	26.76%			
	More than 5	10,579	73.24%			

Notes: *Hukou* is China's household registration system that classifies citizens based on their place of registration and links them to different access to social welfare, education, employment, and management. In this study, *Hukou* is categorized into urban and rural types; Type of employer is classified by ownership: public sector (state-owned or collective enterprises) and private sector (privately owned enterprises or self-employed business); Level of professional title is grouped into two categories: higher (senior-level titles such as "Senior Engineer") and other (including junior, intermediate, and unspecified ranks).

Regression model results for different dimensions of technical expertise

Table C1. OLS regression results.

Variables	Model 0	Model 1
Educational background (ref. Not computer or software related major)		0.106*** (0.017)
Job type of programming (ref. Others)		0.056* (0.025)
Years of professional programming (ref. Less than 4 years)		0.185*** (0.018)
Gender (ref. female)	0.144*** (0.017)	0.117*** (0.017)
Age	−0.012*** (0.001)	−0.012*** (0.001)
Types of *Hukou* (ref. Rural *Hukou*)	0.059*** (0.016)	0.057*** (0.016)
Marriage status (ref. Never married)	0.164*** (0.020)	0.114*** (0.021)
Number of children	−0.087*** (0.014)	−0.074*** (0.014)
Years of education	0.094*** (0.004)	0.090*** (0.004)
Type of employer (ref. Private sector)	−0.049** (0.016)	−0.056*** (0.016)
Monthly income (ref. Less than 10,000 *yuan* RMB)	0.121*** (0.018)	0.083*** (0.018)
Level of professional title (ref. Other)	−0.002 (0.020)	−0.012 (0.020)
Weekly workhours (ref. No more than 40 hours)	−0.045** (0.016)	−0.055*** (0.016)
Number of projects	0.081*** (0.005)	0.068*** (0.005)
Number of team members (ref. No more than 5)	−0.212*** (0.019)	−0.197*** (0.019)
Constant	1.713*** (0.082)	1.722*** (0.081)
R-squared	0.194	0.203
N	14,445	14,445

Notes: *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; Robust standard errors in parentheses.